BANKING DOMAIN ANALYSIS PROJECT

SQL QUERY OUTPUT DOCUMENT

1. **Determine the total count of loan applications**

   - select count(distinct id) as total_loan_applications from bank_load_data;

Results | Messages

	total_loan_applications
1	38576

2. **Determine the recent month-to-date total count of loan applications**

   -select count(distinct id) as MTD_total_loan_applications from bank_load_data where MONTH(issue_date) = 12 and year(issue_date) = 2021;

Results | Messages

	MTD_total_loan_applications
1	4314

3. **Determine the post-month-to-date total count of loan applications**

   - select count(distinct id) as MTD_total_loan_applications from bank_load_data where MONTH(issue_date) = 11 and year(issue_date) = 2021;

Results | Messages

	PMTD_total_loan_applications
1	4035

4. **<u>Determine the Total Funded Amount month-to-date</u>**

-select sum(loan_amount) as MTD_total_funded_amount from bank_load_data where month(issue_date) = 12 and year(issue_date) = 2021;

Results | Messages

	MTD_total_funded_amount
1	53981425

5. **<u>Determine the total funded amount post month-to-date</u>**

-select sum(loan_amount) as PMTD_total_funded_amount from bank_load_data where month(issue_date) = 11 and year(issue_date) = 2021;

Results | Messages

	PMTD_total_funded_amount
1	47754825

6. **<u>Determine the total amount received month-to-date</u>**

-select sum(total_payment) as MTD_total_amount_recvd from bank_load_data where month(issue_date) = 12 and year(issue_date) = 2021;

Results | Messages

	MTD_total_amount_recvd
1	58074380

7. **<u>Determine the total amount received post month-to-date</u>**

-select sum(total_payment) as PMTD_total_amount_recvd from bank_load_data where month(issue_date) = 11 and year(issue_date) = 2021;

Results | Messages

	PMTD_total_amount_recvd
1	50132030

8. **<u>Determine the average interest rate month-to-date</u>**

-select round(avg(int_rate)*100,3) as MTD_Avg_Interest_rate from bank_load_data where month(issue_date) = 12 and year(issue_date) = 2021;

Results | Messages

	MTD_Avg_Interest_rate
1	12.356

9. **<u>Determine the average interest rate post-month-to-date</u>**

-select round(avg(int_rate)*100,3) as PMTD_Avg_Interest_rate from bank_load_data where month(issue_date) = 11 and year(issue_date) = 2021;

Results | Messages

	PMTD_Avg_Interest_rate
1	11.942

10. <u>Determine the average debt-to-income ratio month-to-date</u>

```
-select round(avg(dti),3)*100 as MTD_Avg_DTI_rate from bank_load_data
where MONTH(issue_date) = 12 and year(issue_date) = 2021;
```

Results | Messages

	MTD_Avg_DTI_rate
1	13.7

11. <u>Determine the average debt-to-income ratio post-month-to-date</u>

```
-select round(avg(dti),3)*100 as PMTD_Avg_DTI_rate from bank_load_data
where MONTH(issue_date) = 11 and year(issue_date) = 2021;
```

Results | Messages

	PMTD_Avg_DTI_rate
1	13.3

12. <u>Determine the good loan application percentage</u>

```
-select(count(case when loan_status = 'Fully Paid' or loan_status = 'Current' then id
end)*100)/
        count(id) as good_loan_percentage from bank_load_data;
```

Results | Messages

	good_loan_percentage
1	86

13. <u>Determine the number of good loan applications</u>

```
-select count(id) as total_good_loans from bank_load_data
where loan_status = 'Fully Paid' or loan_status = 'Current'
```

Results | Messages

	total_good_loans
1	33243

14. <u>Determine the total good loan funded amount</u>

```
-select sum(loan_amount) as total_good_loan_amount from bank_load_data
where loan_status = 'Fully Paid' or loan_status = 'Current';
```

Results | Messages

	total_good_loan_amount
1	370224850

15. <u>Determine the total good loan received amount</u>

```
-select sum(total_payment) as total_good_loan_received from bank_load_data
where loan_status = 'Fully Paid' or loan_status = 'Current';
```

Results | Messages

	total_good_loan_received
1	435786170

16. <u>Determine the bad loan application percentage</u>

```
select(count(case when loan_status = 'Charged Off' then id end)*100.0)/
        count(id) as bad_loan_percentage from bank_load_data;
```

100 % No issues fou

Results | Messages

	bad_loan_percentage
1	13.824657818332

17. <u>Determine count of the total bad loans</u>

```
select count(id) as total_bad_loans from bank_load_data
where loan_status = 'Charged Off'
```

Results | Messages

	total_bad_loans
1	5333

18. <u>Determine the total bad loan funded amount</u>

```
-select sum(loan_amount) as total_bad_loan_amount from bank_load_data
where loan_status = 'Charged Off'
```

Results | Messages

	total_bad_loan_amount
1	65532225

19. <u>Determine the total amount of bad loans received amount</u>

```
select sum(total_payment) as bad_loan_received_amount from bank_load_data
where loan_status = 'Charged Off';
```

Results | Messages

	bad_loan_received_amount
1	37284763

20. <u>Loan status grid - 1</u>

```
Select
Loan_status,
count(id) as Total_Loan_Applications,
sum(total_payment) as total_amount_received,
sum(loan_amount) as total_loan_amount,
round(avg(int_rate * 100),3) as average_interest_rate,
round(avg(dti * 100),3) as DTI
from bank_load_data
group by loan_status;
```

Results | Messages

	loan_status	totalLoanCount	total_amount_received	total_loan_amount	average_interest_rate	DTI
1	Fully Paid	32145	411586256	351358350	11.641	13.167
2	Charged Off	5333	37284763	65532225	13.879	14.005
3	Current	1098	24199914	18866500	15.099	14.724

21. <u>Loan status grid - 2</u>

```
select loan_status,
sum(total_payment) as total_amount_received,
sum(loan_amount) as total_loan_amount
from bank_load_data where month(issue_date) = 12 and year(issue_date) = 2021
group by loan_status;
```

Results | Messages

	loan_status	total_amount_received	total_loan_amount
1	Fully Paid	47815851	41302025
2	Charged Off	5324211	8732775
3	Current	4934318	3946625

22. <u>Monthly trends by issue date</u>

```
select month(issue_date) as month_num,datename(month,issue_date) as month_name,
count(id) as total_loan_applications,sum(loan_amount) as Total_Funded_Amount,
sum(total_payment) as total_received_amount from bank_load_data group by
month(issue_date),datename(month,issue_date) order by month(issue_date);
```

Results | Messages

	month_num	month_name	total_loan_applications	Total_Funded_Amount	total_received_amount
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

23. <u>Regional Analysis by state</u>

select address_state, count(id) as total_loan_applications, sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as total_received_amount
from bank_load_data group by address_state order by sum(loan_amount) desc;

Results | Messages

	address_state	total_loan_applications	Total_Funded_Amount	total_received_amount
1	CA	6894	78484125	83901234
2	NY	3701	42077050	46108181
3	TX	2664	31236650	34392715
4	FL	2773	30046125	31601905
5	NJ	1822	21657475	23425159
6	IL	1486	17124225	18875941
7	VA	1375	15982650	17711443
8	PA	1482	15826525	17462908
9	GA	1355	15480325	16728040
10	MA	1310	15051000	16676279
11	OH	1188	12991375	14330148
12	MD	1027	11911400	12985170
13	AZ	833	9206000	10041986
14	CO	770	8976000	9845810

24. <u>Loan term analysis</u>

select term, count(id) as total_loan_applications, sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as total_received_amount from bank_load_data group by term order by sum(loan_amount) desc;

Results | Messages

	term	total_loan_applications	Total_Funded_Amount	total_received_amount
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

25. <u>Employee length analysis</u>

select emp_length, count(id) as total_loan_applications, sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as total_received_amount from bank_load_data group by emp_length order by sum(loan_amount) desc;

Results | Messages

	emp_length	total_loan_applications	Total_Funded_Amount	total_received_amount
1	10+ years	8870	116115950	125871616
2	2 years	4382	44967975	49206961
3	< 1 year	4575	44210625	47545011
4	3 years	4088	43937850	47551832
5	4 years	3428	37600375	40964850
6	5 years	3273	36973625	40397571
7	1 year	3229	32883125	35498348
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

26. <u>Loan Purpose Breakdown</u>

select purpose, count(id) as total_loan_applications,sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as total_received_amount from bank_load_data group by purpose order by sum(loan_amount) desc;

100 % No issues found

Results Messages

	purpose	total_loan_applications	Total_Funded_Amount	total_received_amount
1	Debt consolidation	18214	232459675	253801871
2	credit card	4998	58885175	65214084
3	home improvement	2876	33350775	36380930
4	other	3824	31155750	33289676
5	small business	1776	24123100	23814817
6	major purchase	2110	17251600	18676927
7	car	1497	10223575	11324914
8	wedding	928	9225800	10266856
9	medical	667	5533225	5851372
10	house	366	4824925	5185538
11	moving	559	3748125	3999899
12	educational	315	2161650	2248380
13	vacation	352	1967950	2116738
14	renewable_energy	94	845750	898931

27. <u>Home ownership analysis</u>

select home_ownership, count(id) as total_loan_applications, sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as total_received_amount from bank_load_data group by home_ownership order by sum(loan_amount) desc;

100 % No issues found

Results Messages

	home_ownership	total_loan_applications	Total_Funded_Amount	total_received_amount
1	MORTGAGE	17198	219329150	238474438
2	RENT	18439	185768475	201823056
3	OWN	2838	29597675	31729129
4	OTHER	98	1044975	1025257
5	NONE	3	16800	19053